

J. LUNA ZAGORAC

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TSI Postdoctoral Fellow ◊ McGill University ◊ Montréal, QC

PROFILE

I'm a cosmologist through and through: passionate not just about what our silly little Universe is up to, but also the ways we as humans interact with and understand it. I love to work in radically interdisciplinary ways, from marrying quantum-inspired techniques and numerical simulations of wave dark matter to developing a Python package to map ancient Egyptian star data from hieroglyphs to virtual skies. I'm always keen to share my work & interests through scicomm, teaching, and mentoring.

EMPLOYMENT

McGill University, Montréal, QC Trottier Space Institute Postdoctoral Fellow	<i>Sep 2025 - Aug 2028</i>
Perimeter Institute for Theoretical Physics, Waterloo, ON Postdoctoral Fellow	<i>Sep 2022 - Aug 2025</i>

EDUCATION

Yale University, New Haven, CT Ph.D. in Physics	<i>Aug 2016 - May 2022</i>
Colgate University, Hamilton, NY B.A. with Honors in Astronomy/Physics & Anthropology	<i>Aug 2012 - May 2016</i>

SKILLS AND QUALIFICATIONS

Programming Languages	Python, C/C++, MATLAB, Chapel, Pascal
Python Packages	Jupyter, Matplotlib, Numba, NumPy, SciPy, PyFFTW, AstroPy
Software & Tools	LaTeX, Excel, Mathematica, ImageJ
Communication skills	Science & grant writing, outreach, public speaking, data visualization
Certifications	Yale Poorvu Center Certificate for Public Communication, Certificate of College Teaching Preparation
Languages	English & Serbian (native) French, Italian & Arabic (limited proficiency) Latin & Middle/Late Egyptian (intermediate)

HONORS & AWARDS

Marie Skłodowska-Curie Postdoctoral Fellowship Seal of Excellence <i>Proposal titled "Making Waves with Particles: Towards Scalable Hybrid Simulations of UltraLight Dark Matter" recognized as a high-quality project proposal in a highly competitive evaluation process.</i>	<i>Feb 2025</i>
Leigh Page Award for Excellence in Graduate Student Teaching <i>Award for \$500 which recognizes broad and valuable contributions to physics education at Yale, science communication, and work fostering a welcoming learning environment for students.</i>	<i>Nov 2021</i>
Future Investigator in NASA Earth and Space Science and Technology <i>NASA Grant for \$90k funding two years of doctoral work and independent investigations of ULDM.</i>	<i>May 2020</i>
Loyde and William C. G. Ortel Fellowship in Physics <i>Awarded to an outstanding student pursuing a Ph.D. in Physics.</i>	<i>Nov 2020</i>
Franke Science & Humanities Interdisciplinary Research Award <i>Yale Fellowship for \$3000 funding two years of interdisciplinary work on Egyptian constellations.</i>	<i>Sep 2019</i>
Colgate Physics and Astronomy Department Founders Award <i>Awarded periodically to a senior who has demonstrated four years of outstanding progress and development of their understanding of physics or astronomy.</i>	<i>Apr 2016</i>
Sigma Pi Sigma Physics Honors Society <i>Honorary membership to Sigma Pi Sigma Honors Society.</i>	<i>Apr 2016</i>

10. Zhang, Wang, **Zagorac**, and Easter. “Stone Skipping Black Holes in Ultralight Dark Matter Solitons.” *arXiv: 2602.11512*.
9. **Zagorac** and Symons. “Modelling the Observational Method Behind Ramesside Star Clocks: Contextualizing Relevant Factors.” *Submitted to JHA*.
8. Mirasola, Musoke, Neyrinck, Prescod-Weinstein, and **Zagorac**.^α “The three phases of self-gravitating scalar field ground states.” *arXiv: 2410.02663, submitted to PRD*.
7. Polzin et al. (including **Zagorac**). “Astronomy as a Field: A Guide for Aspiring Astrophysicists.” *arXiv: 2312.04041, submitted to BAAS*.
6. Robles, **Zagorac**, and Padmanabhan. “Scalar Field Dark Matter: Impact of Supernovae-driven blowouts in the central densities of dwarf galaxies.” *MNRAS, 532(2):1980–1990, August 2024*.
5. Gosenca, Eberhardt, Wang, Eggemeier, Kendall, **Zagorac**, and Easter. “Multifield Ultralight Dark Matter.” *Physical Review D 107.8 (2023): 083014*.
4. **Zagorac**, Kendall, Padmanabhan, and Easter. “Soliton Formation and the Core-Halo Mass Relation for Synthetic ULDM Halos: An Eigenstate Perspective.” *Physical Review D 107.8 (2023): 083513*.
3. **Zagorac**, Sands, Padmanabhan, and Easter. “Schrödinger-Poisson Solitons: Perturbation Theory.” *Physical Review D 105.10 (2022): 103506*.
2. Padmanabhan, Ronaghan, **Zagorac**, and Easter. “Simulating Ultralight Dark Matter with Chapel: An Experience Report.” (2019). *2020 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*
1. **Zagorac**, Easter, and Padmanabhan. “GUT-scale primordial black holes: mergers and gravitational waves.” *Journal of Cosmology and Astroparticle Physics 2019.06 (2019): 052*.

INVITED SEMINARS AND COLLOQUIA

* = Virtual

Carleton University Physics Seminar	Mar 2026
TSI Monday Lunch Seminar	Mar 2026
McGill HEP Journal Club	Oct 2025
George Mason University (Colloquium*)	Sep 2025
Bates College (Colloquium)	Dec 2024
Bard College (Colloquium)	Nov 2024
Kavli IPMU	Nov 2024
McMaster University (Colloquium)	Sep 2024
University of Oxford	Apr 2024
American Museum of Natural History	Feb 2024
Canadian Institute for Theoretical Astrophysics	Oct 2023
University of Southern California	Sep 2023
Jet Propulsion Laboratory	Sep 2023
Colgate University (Colloquium)	Apr 2022
Stockholm University*	Jan 2022
Perimeter Institute*	Dec 2021
Stony Brook University*	Nov 2021
University College London*	Oct 2021
Carnegie Observatories*	Oct 2021
University of Hawaii Institute of Astronomy*	Oct 2021
Northwestern University CIERA*	Oct 2021
Newcastle University*	Sep 2021
Center for Computational Astrophysics*	May 2020

Ancient Egyptian Astronomy Meeting	June 2026
<i>Digital Skies for Ancient Contexts:</i>	<i>Royal Astronomical Society</i>
<i>Using Python to Investigate Stellar Data from Ramesside Star Clocks</i>	<i>London, UK</i>
Axions in Japan	Nov 2025
[†] <i>Not Really Quantum Cosmology:</i>	<i>Kavli IPMU</i>
<i>How far can we get by treating a DM halo like an atom?</i>	<i>Tokyo, Japan</i>
American Research Center in Egypt Annual Meeting	Apr 2025
<i>On the Observational Method Behind Ramesside Star Clocks:</i>	<i>San Francisco, CA</i>
<i>Contextualizing Relevant Factors</i>	
Kashiwa-no-ha Dark Matter and Cosmology Symposium	Nov 2024
<i>On ground states of self-gravitating scalar fields</i>	<i>Tokyo, Japan</i>
COSMO'24	Oct 2024
<i>Not Really Quantum Cosmology:</i>	<i>Kyoto University</i>
<i>How far can we get by treating a DM halo like an atom?</i>	<i>Japan</i>
Defining New Simulation Frontiers for Dark Matter Discovery	Oct 2024
<i>An UltraShort Introduction to UltraLight Dark Matter</i>	<i>Toronto, ON</i>
[†] Cosmic Signals of Dark Matter Physics: New Synergies	Jun 2024
<i>Not Really Quantum Cosmology:</i>	<i>Kavli Institute for Theoretical Physics</i>
<i>How far can we get by treating a DM halo like an atom?</i>	<i>Santa Barbara, CA</i>
[†] Canadian Association of Physicists Congress	May 2024
<i>Keynote for Symposium on "Computational Advances in Astrophysics and Cosmology"</i>	<i>London, ON</i>
Egyptian Cultural Heritage Now	Nov 2023
<i>Using Python to Investigate Stellar Data from Ramesside Star Clocks</i>	<i>Cairo, Egypt</i>
APS April Meeting	Apr 2023
<i>Ultralight Dark Matter Dynamics in the Language of Eigenstates</i>	<i>Minneapolis, MN</i>
Testing Gravity 2023	Jan 2023
<i>UltraLight Dark Matter Dynamics in the Language of Eigenstates</i>	<i>Vancouver, BC</i>
240th Meeting of the American Astronomical Society	Jun 2022
<i>A Light in the Dark: UltraLight Dark Matter Phenomenology in Simulations</i>	<i>Virtual</i>
Chapel Implementers and Users Workshop	Jun 2022
<i>UltraLight Dark Matter in Simulations: A Chapel-Powered Eigenstate Perspective</i>	<i>Virtual</i>
American Research Center in Egypt Annual Meeting	Apr 2021
<i>In Search of Lost Time: An Astronomical View of Ancient Egyptian Star Clocks</i>	<i>Virtual</i>
Aspen Winter Conference, A Rainbow of Dark Sectors	Mar 2021
<i>UltraLight Dark Matter & Its Eigenstates</i>	<i>Virtual</i>
[†] Connecticut Digital Humanities	Feb 2021
<i>In Search of Lost Time: An Astronomical View of Ancient Egyptian Star Clocks</i>	<i>Virtual</i>
236th Meeting of the American Astronomical Society	Jun 2020
<i>Parametrizing UltraLight Dark Matter Haloes Through Binary Soliton Core Mergers</i>	<i>Virtual</i>
235th Meeting of the American Astronomical Society	Jan 2020
<i>A Light in the Dark: Ultra Light Dark Matter in Theory and Simulation</i>	<i>Hawaii Convention Center</i>
Great Lakes Cosmology Workshop	Aug 2019
<i>Pseudo-Spectral Solvers for Fuzzy Dark Matter</i>	<i>Rochester Institute of Technology</i>
Tri-Institute Summer School on Elementary Particles	Jul 2018
<i>Gravitational Wave Spectrum of Ultralight Primordial Black Holes</i>	<i>Perimeter Institute</i>

POPULAR SCIENCE PRESENTATIONS

Smart Sips Montréal <i>Cosmic Clues from Ancient Egypt</i>	Jul 2026 Montréal, QC
Astronomy on Tap Montréal <i>A Bestiary of Dark Matter Candidates</i>	Mar 2026 Montréal, QC
TSI & Physics Public Talk <i>Digital Skies for Ancient Contexts:</i> <i>Using Python to Investigate Stellar Data from Ramesside Star Clocks</i>	Dec 2025 McGill University
†EinsteinPlus Workshop for Teachers <i>A Bestiary of Dark Matter Candidates</i>	Jul 2025 Perimeter Institute
†Mexborough & Swinton Astronomical Society <i>Digital Skies for Ancient Contexts:</i> <i>Using Python to Investigate Stellar Data from Ramesside Star Clocks</i>	May 2025 Virtual
†Astronomical Society of Edinburgh <i>Digital Skies for Ancient Contexts:</i> <i>Using Python to Investigate Stellar Data from Ramesside Star Clocks</i>	June 2024 Virtual
†SciComm Collider 2 <i>Discussion leader for “Science Communications and the Humanities”</i>	May 2024 Perimeter Institute
†iTelescope Webinar <i>What we Can’t See in the Universe (and Why it Might Be Fuzzy)</i>	Dec 2023 Virtual
Astronomy on Tap Kitchener-Waterloo <i>A Bestiary of Dark Matter Candidates</i>	Dec 2023 Kitchener, ON
†Canadian Undergraduate Physics Conference Panelist on <i>Change Your Basis: From Expert to Public</i>	Oct 2023 University of Waterloo
†Royal Astronomical Society of Canada Mississauga Centre Speaker Night <i>What we Can’t See in the Universe (and Why it Might Be Fuzzy)</i>	Oct 2023 Mississauga, ON
†SciComm Collider <i>A Bestiary of Dark Matter Candidates</i>	Apr 2023 Perimeter Institute
†David Dunlap Observatory Speaker’s Night <i>What we Can’t See in the Universe (and Why it Might Be Fuzzy)</i>	Dec 2022 Virtual
†Yale Institute of Sacred Music <i>Cosmogonies, Cosmologies, & Time</i>	Mar 2021 Virtual
Bay Area Science Festival “Astro Coffee” <i>Cosmic Archaeology, or: How Do We Know the Things We Know?</i>	Oct 2020 Virtual
Ask a Scientist Webinar <i>Dark Matter</i>	May 2020 Virtual
Astronomy on Tap New Haven <i>Cosmic Archaeology, or: How Do We Know the Things We Know?</i>	Sep 2019 New Haven, CT
Yale 3 Minute Thesis Competition Finalist <i>How Small Black Holes Teach Us about the Big Bang</i>	Apr 2019 Yale University

SCIENCE COMMUNICATION

Public Communication Certificate

Oct 2021

Yale Poorvu Center for Teaching and Learning

Certification in public communication through extensive preparation for the 3MT Competition. Areas: Text, Speech, and Visual Design; Feedback and Revision; Peer and Interdisciplinary Collaboration.

Outreach Volunteering

Mentor, McGill Hackathon

Feb 2026

STEM Pen Pal, Letters to a Pre-Scientist

Sep 2024 - Present

Volunteer, Ask a Scientist at Perimeter Institute

July 2025

Volunteer, Explore Galileo Exhibit at Perimeter Institute

Feb 2024

Volunteer, Dark Matter Night at Perimeter Institute

Oct 2022

Astronomy Ambassador, American Astronomical Society

Jan 2020 - 2022

Volunteer, Yale Pathways to Science

Fall 2018 - Spring 2019

Activity Leader, CT Students Exploring Engineering Day

Spring 2018

Activity Leader, Girls Science Investigations

Sep 2016 - Mar 2020

Writing

FirstPrinciples Contributing Author

May 2024 - Present

Astrobites Media Intern at AAS238

Jun 2021

Astrobites Contributing Author (>20 [articles and interviews](#))

Dec 2019 - Dec 2021

ComSciCon at the American Institute of Physics Participant

Sep 2019

TEACHING EXPERIENCE

Certificate of College Teaching Preparation (CCTP)

May 2022

Yale Poorvu Center for Teaching and Learning

A record of participation in teaching activities and reflections on those experiences. Earning the CCTP, also meets requirements for the [Center for Integration of Research, Teaching and Learning](#) Associate.

Curriculum Development & Lecturing

SIRUS B Virtual Events for Remote Gathering and Engagement

Jan 2024

Lecture on “Cosmology and Dark Matter” as part of programming for Hawaiian middle school girls.

Tri-Institute Summer School on Elementary Particles (TRISEP)

June 2023

Lecture on “Axion-Like Particles (and Why We Love Them).”

The Yale Summer Program in Astrophysics

Jul 2022

Lecture on “Comparing Cosmologies: the History of the Cosmos from Pyramids to Space Telescopes.”

Yale Institute of Sacred Music

Mar 2021

Lecture in graduate-level religion class on “Cosmogonies, Cosmologies, & Time”

Yale Bootcamp on Physics Fundamentals

Summer 2019 - 2021

Co-developed a curriculum for 20 hours of Classical Mechanics instruction, met weekly with staff supervisor to polish lectures and example problems. Delivered 10 hours of lecture at the Bootcamp. Developed a Mathematica tutorial for incoming graduate students. Re-vamped the curriculum and moved it online for Summer 2020 and Summer 2021.

Teaching Fellow Positions

S&DS176 - YData: Humanities Data Mining	<i>Spring 2022</i>
PHYS/ASTR600 - Cosmology	<i>Fall 2020</i>
PHYS442 - Introduction to Nuclear and Elementary Particle Physics	<i>Spring 2020</i>
PHYS410 - Classical Mechanics	<i>Fall 2019</i>
ASTR343 - Gravity, Astrophysics, and Cosmology	<i>Spring 2019</i>
PHYS170/171 - University Physics for the Life Sciences	<i>Fall 2017 - Spring 2018</i>
PHYS165/166 - General Physics Laboratory	<i>Fall 2016 - Spring 2017</i>

SERVICE & LEADERSHIP

Committee Work

TSI Summer Undergrad Research Awards Committee	<i>Mar 2026</i>
Perimeter Institute Anti-Racism Working Group Member	<i>Sep 2022 - Aug 2025</i>
Astrobites Diversity, Equity, and Inclusion Committee Member	<i>Mar 2020 - Dec 2021</i>
Physics Climate and Diversity Committee Member	<i>Jan 2018 - May 2020</i>

Conference & Seminar Organization

Co-organizer: Black in Physics Week at Yale Event Series	2020
Volunteer: Conference for Undergraduate Women in Physics	2019-2020
Co-organizer: Equity in the Job Search Symposium	2018-2019

University Positions

Yale Digital Humanities Lab Consultant	<i>Sep 2020 - May 2022</i>
McDougal Graduate Student Life Fellow at Yale	<i>Aug 2018 - May 2019</i>
Graduate Affiliate, Pauli Murray College at Yale	<i>Fall 2017 - Spring 2022</i>

MENTORSHIP

Perimeter Institute PSI Start Summer Undergraduate Research Project	2024
<i>Dynamical Heating in Early Fuzzy Galaxies</i>	<i>with Prof. Katie Mack</i>
Nikki Veilleux (Bishop's University)	

Perimeter Institute PSI Winter School Masters Research Project	2023
<i>Dynamical Heating in Early Fuzzy Galaxies</i>	<i>with Prof. Katie Mack</i>
Cole Coughlin (Ph.D. at PI starting 2023)	
Anna Knörr (Ph.D at Harvard starting 2024)	

Perimeter Institute Postoc-PhD Mentoring

Alice Chen, Samantha Hergott	2023-2025
Maxence Corman, Ramiro Cayuso	2022-2023

Yale Undergraduate Researchers Supervised

Claire Recamier (Researcher at Los Alamos National Lab starting 2023): <i>Stellar Streams in UltraLight Dark Matter Halos</i>	<i>Jun 2021 - May 2023</i>
Isabel Sands (Ph.D. at Caltech starting 2021): <i>Constructing a Binary Soliton Merger Library</i> <i>Linear Approximations to UltraLight Dark Matter Stationary States</i>	<i>Jan 2020 - Jun 2021</i>

Other Formalized Mentoring Activities

SU(5) Group Mentor

Fall 2020

Científico Latino Graduate Student Mentoring Initiative (GSMI) Mentor

Fall 2019

Women in Science at Yale (WISAY) Mentor

2016-2019